GCSE Mathematics practice paper

Answer all questions. Show your working.

Question 1 (2 marks)

Simplify $x^5 \times x^3$.

Answer: x^8

Question 2 (3 marks)

Solve the simultaneous equations $2x + y = 11$ and $x - y = 1$.

Answer: $x = 4, y = 3$

AI-generated worked solutions - may contain mistakes

Question 3 (2 marks)

Work out $\frac{3}{4} + \frac{1}{8}$. Give your answer as a fraction.

Answer: $7/8$

Question 4 (3 marks)

A coat costs £80. In a sale the price is reduced by 15%. Work out the sale price.

Answer: £ 68

(Not checked by calculation: look over this one)

Question 5 (2 marks)

Write 0.000 52 in standard form.

Answer: 5.2×10^{-4}

AI-generated worked solutions

Written by an AI and may contain mistakes. Answers marked as not confidently solved or not checked need a careful look.

Question 1 (2 marks)

Simplify x^5 × x^3.

Checked by calculation

When multiplying powers with the same base, add the exponents.

$x^5 \times x^3 = x^{5+3}$

Calculate the sum of the exponents.

$5 + 3 = 8$

Answer: x^8

To simplify the expression, add the indices (exponents) of the like bases according to the laws of indices.

Question 2 (3 marks)

Solve the simultaneous equations 2x + y = 11 and x - y = 1.

Checked by calculation

Add the two equations together to eliminate y

$(2x + y) + (x - y) = 11 + 1$

Simplify and solve for x

$3x = 12 \Rightarrow x = 4$

Substitute x = 4 into the second equation

$4 - y = 1$

Rearrange to find y

$y = 4 - 1 = 3$

Answer: $x = 4, y = 3$

Add the equations to eliminate y, solve for x, then substitute back to find y.

Question 3 (2 marks)

Work out 3/4 + 1/8. Give your answer as a fraction.

Checked by calculation (against the AI's reading of the question)

Find the common denominator for 4 and 8, which is 8.

$\frac{3}{4} = \frac{6}{8}$

Now add the two fractions together.

$\frac{6}{8} + \frac{1}{8} = \frac{7}{8}$

Answer: $7/8$

To add the fractions, convert 3/4 to have a denominator of 8, then add the numerators.

Question 4 (3 marks)

\nA coat costs £80. In a sale the price is reduced by 15%. Work out the sale price.

Not checked by calculation: look over this one

Find 15% of 80.

$$15\% \text{ of } 80 = \frac{15}{100} \times 80 = 12$$

Subtract this discount from the original price.

$$80 - 12 = 68$$

Answer: £ 68

To find the sale price, calculate 15% of £80 and subtract it from the original cost.

Question 5 (2 marks)

Write 0.000 52 in standard form.

Checked by calculation

Move the decimal point to the right until there is one non-zero digit before it.

$$0.00052 \to 5.2$$

Count the number of places moved, which is 4. The decimal moved 4 places.

Since we moved right, the exponent is negative.

$$5.2 \times 10^{-4}$$

Answer: 5.2×10^{-4}

To write a number in standard form, move the decimal point so that there is one non-zero digit to its left and multiply by 10 raised to the power of the number of places moved.